

BANARAS HINDU UNIVERSITY

B.Com. (Hons.) — Semester IV Examination, 2015-16

Paper No. BCH-224: Business Mathematics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. All questions carry equal marks (14 marks each).***Question 1.**(a) Find the sum of the first n odd natural numbers. [7](b) If a, b, c are in A.P., then prove that:

$$\frac{1}{\sqrt{b} + \sqrt{c}}, \frac{1}{\sqrt{c} + \sqrt{a}}, \frac{1}{\sqrt{a} + \sqrt{b}}$$

are also in A.P. [7]

OR(a) The p^{th} and q^{th} terms of an A.P. are c and d respectively. Find the r^{th} term. [7](b) The sum of n terms of an A.P. is 40, the common difference is 2, and the last term is 13. Find the number of terms. [7]**Question 2.**

(a) The sum of three numbers in A.P. is 21. If the second number is reduced by 1 and the third number is increased by 1, then the numbers obtained are in G.P. Find the numbers. [7]

(b) If a, b, c are in G.P. and x, y are the arithmetic means between a, b and b, c respectively, prove that:

$$\frac{a}{x} + \frac{c}{y} = 2$$

[7]

OR

(a) If the arithmetic mean of two numbers is 10 and their geometric mean is 6, then find the numbers. [7]

(b) If $\frac{1}{x+y}, \frac{1}{2y}, \frac{1}{y+z}$ are in A.P., then prove that x, y, z are in G.P. [7]

Question 3.

- (a) If ${}^{2n+1}P_{n-1} : {}^{2n-1}P_n = 3 : 5$, then find the value of n . [7]
- (b) How many numbers between 100 and 10,000 can be made with the 8 digits 1, 2, 3, 0, 4, 5, 6, 7? (Repetition of digits is not allowed.) [7]

OR

- (a) How many triangles can be formed by joining the angular points of a decagon? How many diagonals does it have? [7]
- (b) A box contains 7 red, 6 white, and 4 blue balls. Find how many selections of three balls can be made so that none of them is red. [7]
-

Question 4.

- (a) Find the term independent of x in the expansion of $\left(x^2 + \frac{1}{x}\right)^{12}$. [7]
- (b) In the expansion of $(1+x)^{43}$, the coefficients of the $(2r+1)^{\text{th}}$ term and $(r+2)^{\text{th}}$ term are equal. Find the value of r . [7]

OR

- (a) Find the 19^{th} term of $\left(2x^{1/2} - y^{1/3}\right)^{20}$. [7]
- (b) Find the coefficient of x^{15} in the expansion of $(x - x^2)^{10}$. [7]
-

Question 5.

- (a) If $f(x) = \log\left(\frac{1-x}{1+x}\right)$, then prove that:

$$f(a) + f(b) = f\left(\frac{a+b}{1+ab}\right)$$

[4]

- (b) Evaluate:

$$\lim_{x \rightarrow 0} \frac{\sqrt{1+x} - \sqrt{1-x}}{x}$$

[4]

- (c) If $y = x^{x^{x^{\dots\infty}}}$, then prove that:

$$x \frac{dy}{dx} = \frac{y^2}{1 - y \log x}$$

[6]

OR

- (a) Simplify:

$$\int \left(\sqrt{x} + \frac{1}{\sqrt{x}} \right)^2 dx$$

[4]

- (b) Simplify:

$$\int \frac{x}{\sqrt{x+1}} dx$$

[4]

- (c) Simplify:

$$\int \frac{x+1}{(x+2)^2} dx$$

[6]